

Exercise 1. There are 17 students (11 boys and 6 girls) in a class. How many ways are there to form a 7-member committee for the class if:

1. At least 4 members must be boys?
2. At least one of the members must be a specific boy, Omar?

Four offices, Chairperson, Vice-Chairperson, Secretary, and Treasurer are to be assigned within a committee. Determine the number of ways to assign these offices if:

3. The Chairperson must be a girl, and the Vice-Chairperson must be a boy?
4. The Chairperson and the Secretary cannot belong to the same gender?

Solution:

1. At least 4 members must be boys

The class has 11 boys and 6 girls, and the committee consists of 7 members. To satisfy the condition that at least 4 members are boys, we must consider cases where there are 4, 5, 6, or 7 boys in the committee.

Case 1: 4 boys and 3 girls

- Choose 4 boys from 11: $\binom{11}{4}$.
- Choose 3 girls from 6: $\binom{6}{3}$.

Total ways: $\binom{11}{4} \cdot \binom{6}{3}$.

Case 2: 5 boys and 2 girls

- Choose 5 boys from 11: $\binom{11}{5}$.
- Choose 2 girls from 6: $\binom{6}{2}$.

Total ways: $\binom{11}{5} \cdot \binom{6}{2}$.

Case 3: 6 boys and 1 girl

- Choose 6 boys from 11: $\binom{11}{6}$.
- Choose 1 girl from 6: $\binom{6}{1}$.

- Total ways: $\binom{11}{6} \cdot \binom{6}{1}$.

Case 4: 7 boys and 0 girls

- Choose 7 boys from 11: $\binom{11}{7}$.
- No girls are chosen.

Total ways: $\binom{11}{7}$.

By (AP), the total number is

$$\binom{11}{4} \cdot \binom{6}{3} + \binom{11}{5} \cdot \binom{6}{2} + \binom{11}{6} \cdot \binom{6}{1} + \binom{11}{7}. \quad (0.75pt)$$

2. At least one of the members must be a specific boy, Omar

First, we choose Omar in $\binom{1}{1}$ ways. Then we complete the committee by choosing the remaining 6 members from the remaining 16 members in $\binom{16}{6}$. The total number is $\binom{1}{1} \binom{16}{6}$. (0.75pt)

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3. The Chairperson must be a girl, and the Vice-Chairperson must be a boy

- Choose the Chairperson from 6 girls: $\binom{6}{1}$.

- Choose the Vice-Chairperson from 11 boys: $\binom{11}{1}$.

- Choose the remaining 2 of the remaining 15 students in $\binom{15}{2}$ ways and assign offices (Secretary and Treasurer) in $2!$ ways.

The total number is

$$\binom{6}{1} \cdot \binom{11}{1} \cdot 15 \cdot 14. \quad (1pt)$$

4. The Chairperson and Secretary cannot belong to the same gender

There are two cases to consider:

Case 1: Chairperson is a girl, and Secretary is a boy

- Choose the Chairperson from 6 girls: $\binom{6}{1}$.

- Choose the Secretary from 11 boys: $\binom{11}{1}$.

- Assign the remaining 2 offices (Vice-Chairperson and Treasurer) to any 2 of the remaining 15 students in $15 \cdot 14$ ways.

Case 2: Chairperson is a boy, and Secretary is a girl

- Choose the Chairperson from 11 boys: $\binom{11}{1}$.

- Choose the Secretary from 6 girls: $\binom{6}{1}$.

- Assign the remaining 2 offices (Vice-Chairperson and Treasurer) to any 2 of the remaining 15 students in $15 \cdot 14$ ways.

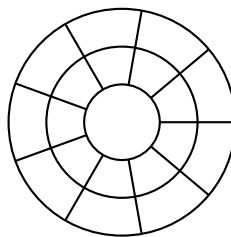
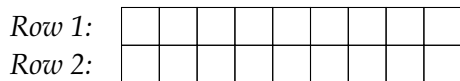
The total number is

$$\left[\binom{6}{1} \cdot \binom{11}{1} + \binom{11}{1} \cdot \binom{6}{1} \right] \cdot 15 \cdot 14. \quad (1pt)$$

Exercise 2. Consider n fathers, each with their child, attending a meeting in a room with $2n$ seats. How many ways can the seats be arranged in **two parallel rows**:

- Without restrictions.
- If all fathers must sit in the same row.
- If no father sits in the same row as his child, but they sit in the same position in their respective rows.
- If each father has only children as neighbors and vice versa.

How many ways can the seats be arranged in **two circular rows** for the first three cases of the previous question?



Solution:

• Two parallel rows:

1. No restrictions:

The number of ways to arrange $2n$ people (fathers and children) in $2n$ seats is simply the number of permutations of $2n$ distinct people. Therefore, the total number of arrangements is:

$$(2n)! \quad (1pt).$$

2. All fathers must sit in the same row:

We need to arrange all n fathers in one row and all n children in the other row.

First, we choose which row the fathers will sit in. There are $\binom{2}{1} = 2$ choices for this. After selecting the row for the fathers, we can arrange the n fathers in that row in $n!$ ways, and the n children can be arranged in the other row in $n!$ ways. Therefore, the total number of arrangements is:

$$2 \times n! \times n! \quad (1.25pt).$$

3. No father sits in the same row as his child, but they sit in the same position in their respective rows:

In this case, each father and his child form a "block" that must sit in corresponding positions in different rows. If a father is sitting in position i in one row, his child must sit in position i in the other row, but not in the same row as the father.

First, we arrange the n father-child blocks in any order, which can be done in $n!$ ways. Since each father and his child must sit in different rows, and the father and child can swap positions within their block, we multiply by a factor of 2^n (as each block has 2 possible internal arrangements). Therefore, the total number of arrangements is:

$$n! \times 2^n \quad (1.25pt).$$

4. Each father has only children as neighbors and vice versa:

In this case, the seating must alternate between fathers and children, ensuring that each father sits next to a child.

First, we fix the arrangement of fathers and children such that they alternate. Since there are n fathers and n children, the seating will follow the pattern: father-child-father-child-..., continuing until all seats are filled.

There are $n!$ ways to arrange the fathers in their designated seats, and $n!$ ways to arrange the children in their designated seats. Additionally, we can swap the positions of the two rows, meaning we can have one row starting with a father and the other starting with a child, or vice versa. This gives an extra factor of 2. Therefore, the total number of arrangements is:

$$2 \times (n!)^2 \quad (1.5pt).$$

• **Two circular rows:**

1. **No restrictions:**

In the case of circular rows, we must account for rotational symmetry. To eliminate this symmetry, we fix one person's position in one of the rows and then arrange the remaining people in the remaining seats.

First, we select the n people who will sit in the first row, which can be done in $\binom{2n}{n}$ ways. To break the rotational symmetry, we fix one person (e.g., a father) in a specific seat, and arrange the remaining $n-1$ people in the same row. This can be done in $(n-1)!$ ways. Then, we permute the n people in the other row, which can be done in $n!$ ways.

Therefore, the total number of arrangements is:

$$\binom{2n}{n} (n-1)! n! = \frac{(2n)!}{n} \quad (1.25pt).$$

2. **All fathers must sit in the same row:**

As in the previous case, we account for the circular symmetry. First, we choose one of the rows for the fathers (2 choices). Then, we fix one father in one of the n seats in the chosen row to eliminate the circular symmetry, and arrange the remaining $n-1$ fathers in the remaining seats in that row. The number of ways to arrange the remaining $n-1$ fathers is $(n-1)!$.

The n children are placed in the other row, which can be done in $n!$ ways.

Therefore, the total number of arrangements is:

$$2 \times (n-1)! \times n! \quad (1.25pt).$$

3. **No father sits in the same row as his child, but they sit in the same position in their respective rows:**

As in the case with parallel rows, we first arrange the n father-child blocks, where each father and his child must sit in corresponding positions in different rows. In a circular arrangement, we fix one block to eliminate the rotational symmetry. This can be done in $(n-1)!$ ways for the remaining $n-1$ blocks.

The factor 2^n arises because each father-child pair can swap seats within their block, and we multiply by this factor for each pair.

Therefore, the total number of arrangements is:

$$(n-1)! \times 2^n \quad (1.25pt).$$

Mid-term exam solution

Exercise 3. I) Consider the linear inequality

$$x_1 + x_2 + \cdots + x_n \leq k,$$

where $x_1, x_2, \dots, x_n, k$ are non-negative integers.

- Find the total number of non-negative integer solutions to the inequality.
- Discuss the number of solutions if $x_2 \geq a$, where a is a positive integer.
- Find the number of solutions if $x_5 + x_6 = b$, where $b \leq k$.
- Find the number of solutions if the inequality is bounded below by c and above by k

$$c \leq x_1 + x_2 + \cdots + x_n \leq k, \quad \text{where } c \text{ is a non-negative integer.}$$

II) 1. Use a combinatorial argument to find and prove a formula for

$$\sum_{r+s+t=m} \binom{n_1}{r} \binom{n_2}{s} \binom{n_3}{t}.$$

- What is the coefficients of $x^{14}y^{46}z^{20}t^{24}$ in the expansion of $(2x^2 - 4y^{23} + 7z - t^8)^{33}$?
- How many non-zero terms are there in the expansion of $(x^2 - y^7 + 13z^5 - t^3 + u)^{33}$?

Solution:

I) (a) The inequality can be rewritten as an equation by introducing a new variable x_{n+1} such that:

$$x_1 + x_2 + \cdots + x_n + x_{n+1} = k.$$

Here, $x_{n+1} \geq 0$. The total number of non-negative integer solutions is the same as the number of solutions to this equation, which is given by:

$$\binom{k+n+1-1}{n+1-1} = \binom{k+n}{n}. \quad (1pt).$$

(b) Number of solutions if $x_2 \geq a$

Substitute $x_2 = x'_2 + a$, where $x'_2 \geq 0$. The equation becomes:

$$x_1 + x'_2 + x_3 + \cdots + x_n \leq k - a.$$

Introduce a new variable x_{n+1} for the inequality:

$$x_1 + x'_2 + x_3 + \cdots + x_n + x_{n+1} = k - a.$$

The number of solutions is:

$$\binom{k-a+n+1-1}{n+1-1} = \binom{k-a+n}{n}, \quad (0.75pt).$$

provided $k \geq a$.

If $k < a$, there are no solutions. (0.25 pt).

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(c) The number of solutions to the equation $x_5 + x_6 = b$ is given by:

$$\binom{b+1}{1}.$$

For each solution of this equation, we need to determine the number of solutions to:

$$x_1 + x_2 + x_3 + x_4 + x_7 + \cdots + x_n \leq k - b.$$

Introducing a new variable x_{n+1} such that:

$$x_1 + x_2 + x_3 + x_4 + x_7 + \cdots + x_n + x_{n+1} = k - b.$$

The number of non-negative integer solutions to this equation is given by:

$$\binom{(k-b) + (n-1)}{n-1},$$

provided $b \leq k$.

By the (MP), the total number of solutions is obtained by combining the number of solutions for $x_5 + x_6 = b$ with the number of solutions for $x_1 + x_2 + x_3 + x_4 + x_7 + \cdots + x_n \leq k - b$. Thus, the total number of solutions is:

$$\binom{b+1}{1} \binom{(k-b) + (n-1)}{n-1}. \quad (1.5pt).$$

(d) Number of solutions if $c \leq x_1 + x_2 + \cdots + x_n \leq k$

Split the inequality into two parts:

$$x_1 + x_2 + \cdots + x_n = s, \quad c \leq s \leq k.$$

For each s , the number of solutions to $x_1 + x_2 + \cdots + x_n = s$ is:

$$\binom{s+n-1}{n-1}.$$

Sum over all possible values of s in the range $c \leq s \leq k$:

$$\sum_{s=c}^k \binom{s+n-1}{n-1}. \quad (1.5pt).$$

(a) We can use a combinatorial argument similar to the Vandermonde identity to prove the given formula.

The formula counts the number of ways to choose a total of m objects from three distinct sets: n_1 objects of type 1, n_2 objects of type 2, n_3 objects of type 3.

Now, we can divide these m objects into three groups:

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- r objects from the first set of n_1 objects
- s objects from the second set of n_2 objects
- t objects from the third set of n_3 objects

such that $r + s + t = m$.

The number of ways to do this is:

$$\binom{n_1}{r} \binom{n_2}{s} \binom{n_3}{t}$$

Summing over all possible values of r, s , and t that satisfy $r + s + t = m$ gives us the total number of ways to choose m objects from the three sets. We can think of this as selecting m objects from a total of $n_1 + n_2 + n_3$ objects.

Consider a set of $n_1 + n_2 + n_3$ objects. We can choose m objects from this set in $\binom{n_1 + n_2 + n_3}{m}$ ways. Therefore, we have:

$$\sum_{r+s+t=m} \binom{n_1}{r} \binom{n_2}{s} \binom{n_3}{t} = \binom{n_1 + n_2 + n_3}{m} \quad (1.25pt).$$

This is the desired formula.

(b) The coefficients of $x^{14}y^{46}z^{20}t^{24}$ in $(2x^2 - 4y^{23} + 7z - t^8)^{32}$. Each term contributes as:

$$\binom{32}{a, b, c, d} (2x^2)^a (-4y^{23})^b (7z)^c (-t^8)^d,$$

where $a + b + c + d = 32$. The exponents of x, y, z, t must match:

$$2a = 14, \quad 23b = 46, \quad c = 20, \quad 8d = 24.$$

Solve for a, b, c, d :

$$a = 7, \quad b = 2, \quad c = 20, \quad d = 3.$$

The coefficient is:

$$\binom{32}{7, 2, 20, 3} \cdot (2^7) \cdot (-4)^2 \cdot 7^{20} \cdot (-1)^3. \quad (0.75pt).$$

(c) Number of non-zero terms in the expansion of $(x^2 - y^7 + 13z^5 - t^3 + u)^{33}$.

Each term in the expansion corresponds to a unique multinomial term:

$$\binom{33}{a, b, c, d, e} x^{2a} (-y^7)^b (13z^5)^c (-t^3)^d u^e,$$

where $a + b + c + d + e = 33$. The term is non-zero as long as each variable has a non-negative integer exponent.

The number of such terms is equal to the number of non-negative integer solutions to $a + b + c + d + e = 33$:

$$\binom{33 + 5 - 1}{5 - 1} = \binom{37}{4}. \quad (0.75pt).$$